

5.7 Divergence Theorem Worksheet

1. Suppose $\mathbf{F} = \langle x^3 + y^3, y^3 + z^3, z^3 + x^3 \rangle$ and σ is the sphere of radius 2 centered at the origin. Use the Divergence Theorem to calculate the flux of $\mathbf{F}$ across σ .

2. Let $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$. Determine the flux across the surface of the unit cube existing entirely within the first octant with outward orientation.

3. Let $\mathbf{F}(x, y, z) = y^2\mathbf{i} + x^2\mathbf{j} + z\mathbf{k}$ and σ be the surface of the solid bounded above by the paraboloid $z = 9 - x^2 - y^2$ and below by the plane $z = 1$. Determine the flux across σ with outward orientation.

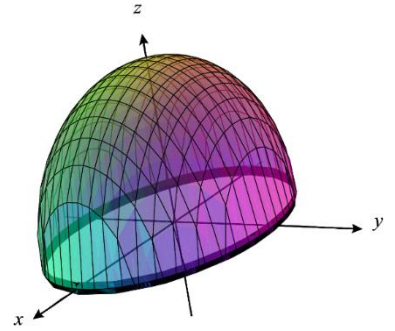
4. Let $\mathbf{F}(x, y, z) = \mathbf{j} + (z + 2)\mathbf{k}$ and consider the solid E formed by the following four surfaces:

- S_0 : given by $z = \sin(y)$ where $0 \leq y \leq \pi$
- S_1 : given by $x = 0$
- S_2 : given by $x = 3$
- S_3 : given by the xy -plane

a) Determine the flux of $\mathbf{F}$ through ∂E with respect to inward normals.

b) The solid E was made from four surfaces as stated above. The flux of $\mathbf{F}$ through exactly two of S_1 , S_2 , and S_3 is exactly 0. Determine which of these corresponds to a nonzero flux.

5. Set up, but do not evaluate, a triple integral which represents the flux of $\mathbf{F}$ across σ where $\mathbf{F}(x, y, z) = \langle x^2 + \sin(y), x + y^2, z + e^x \rangle$ and σ is the surface of the solid within the elliptical cylinder $4x^2 + 9y^2 = 36$ bounded above by the ellipsoid $z = \frac{1}{2}\sqrt{36 - 4x^2 - 9y^2}$ and bounded below by the plane $z = 0$ with outward orientation.



6. Suppose G is a simple solid region in $\mathbb{R}^3$ and let σ be the boundary surface of G , given with positive orientation. Suppose you are told G has a volume of 10. If an object of constant density occupies the region G , then prove that $\frac{1}{20} \iint_{\sigma} \mathbf{B} \cdot d\mathbf{S}$ where $\mathbf{B}(x, y, z) = \langle z, xy, xz \rangle$ is the x -coordinate of the center of mass of the solid.